

17	A hemispherical bowl of radius 200 mm is completely filled with water at rest. If its side is tapped gently, a circular water wave pulse can be produced on the surface of the water which travels inwards with a speed of 250 mm s^{-1}. What is the radius of the pulse and its direction of travel one second after the pulse is produced?	
	A	zero, stationary
	B	50 mm, inwards
	C	50 mm, outwards
	D	100 mm, outwards
	L2	Answer: C After 1 second, the wave would have travelled a total distance of 250 mm. However, since the radius of the bowl is 200 mm, there will be a reflection of the wave at the centre of the bowl when the wave has travelled 200 mm. The wave would have then travelled outward for 50 mm. Thus, the radius of the circle will be 50 mm.

19 A stationary wave is formed on a stretched string of length l between two points as shown